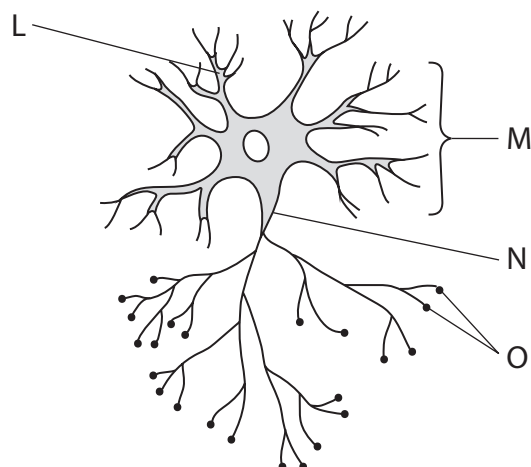


3 The diagram shows a relay neurone.



(a) (i) Complete the table by filling in the names of each labelled part.

(2)

Label	Part
L	
M	
N	
O	

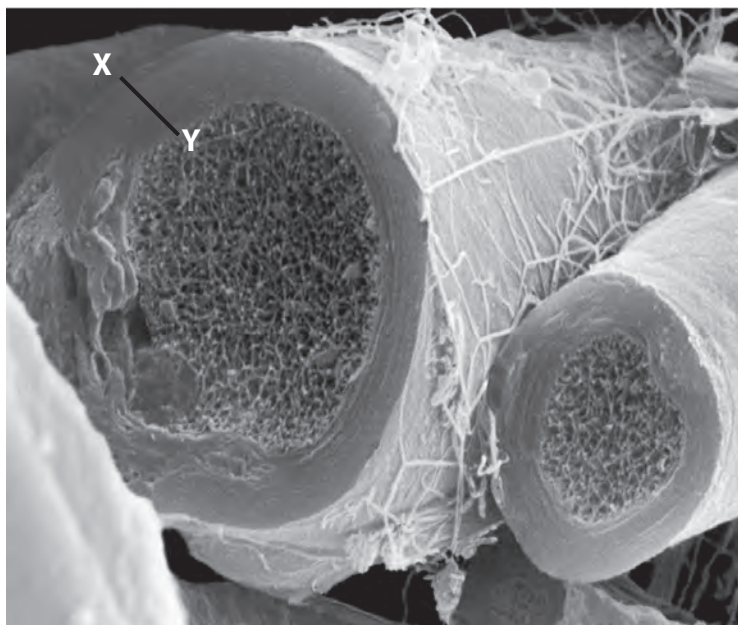
(ii) Lidocaine is a drug that is used as a local anaesthetic.

Lidocaine works by

(1)

- ☐ **A** blocking the binding of acetylcholine to its receptor
- ☐ **B** blocking the release of acetylcholine
- ☐ **C** inhibiting potassium ion channels
- ☐ **D** inhibiting sodium ion channels

(b) The photograph shows a nerve with an outer layer of connective tissue.



(Source: © Science Photo Library/Alamy Stock Photo)

Magnification $\times 3300$

(i) State the type of microscope used to produce this image.

(1)

(ii) Calculate the thickness of the connective tissue layer between the points X and Y.

Give your answer in micrometres (μm).

(2)

Answer μm

(iii) The nerve shown in the photograph contains many myelinated neurones.

Explain why a myelinated neurone conducts an impulse faster than a non-myelinated neurone of the same diameter.

(3)

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(Total for Question 3 = 9 marks)